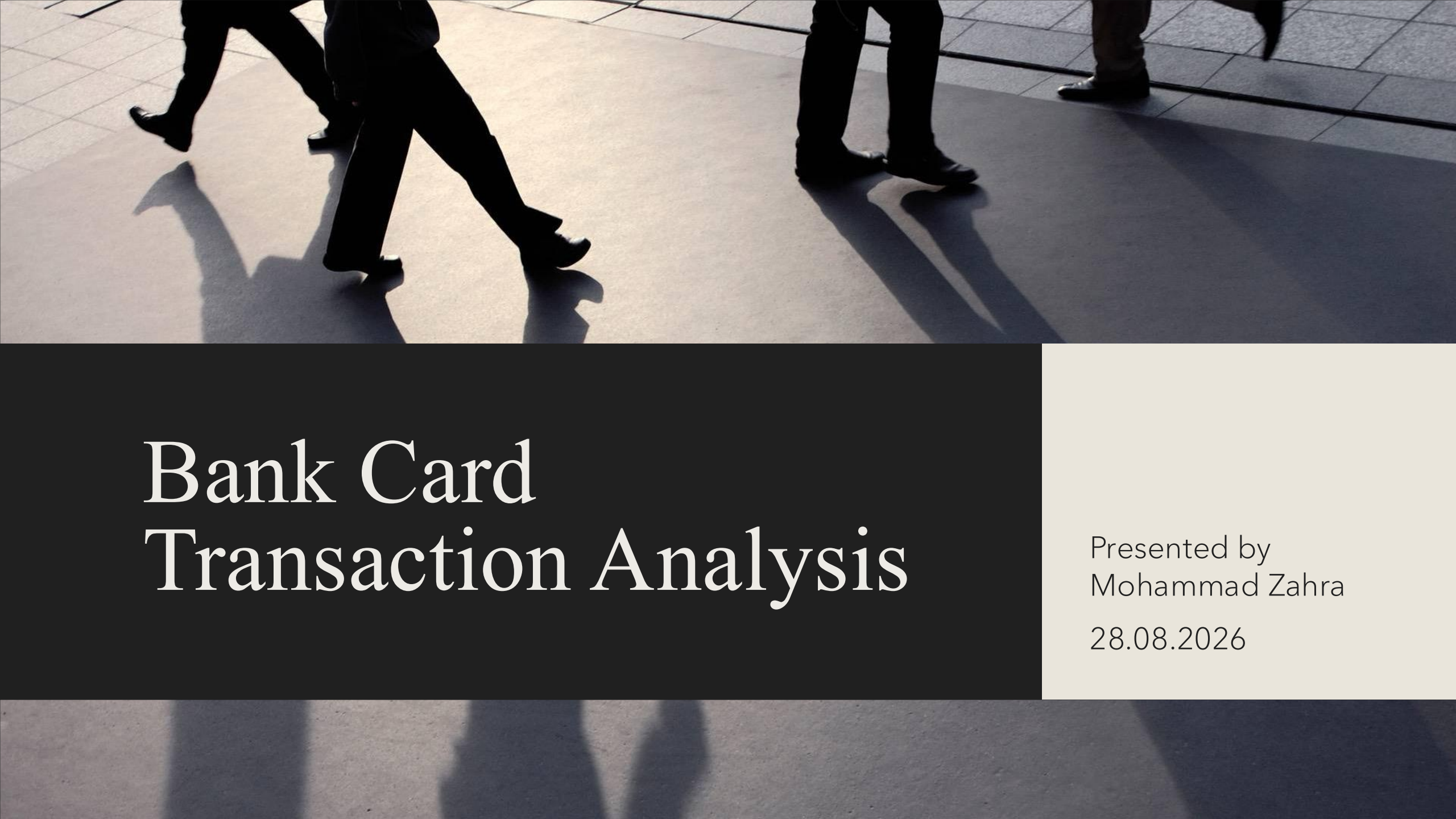




Bank Card Transaction Analysis

Presented by
Mohammad Zahra

28.08.2026

Project Scope & Constraints

In Scope

1. Customers and bank accounts
2. Payment cards
3. Card payment transactions
4. Merchants and merchant categories
5. Transaction locations and time periods

Out of Scope

1. Loans and mortgages
2. Bank transfers and investments
3. Cash withdrawals
4. Fraud detection and credit risk
5. Online banking / other banking products

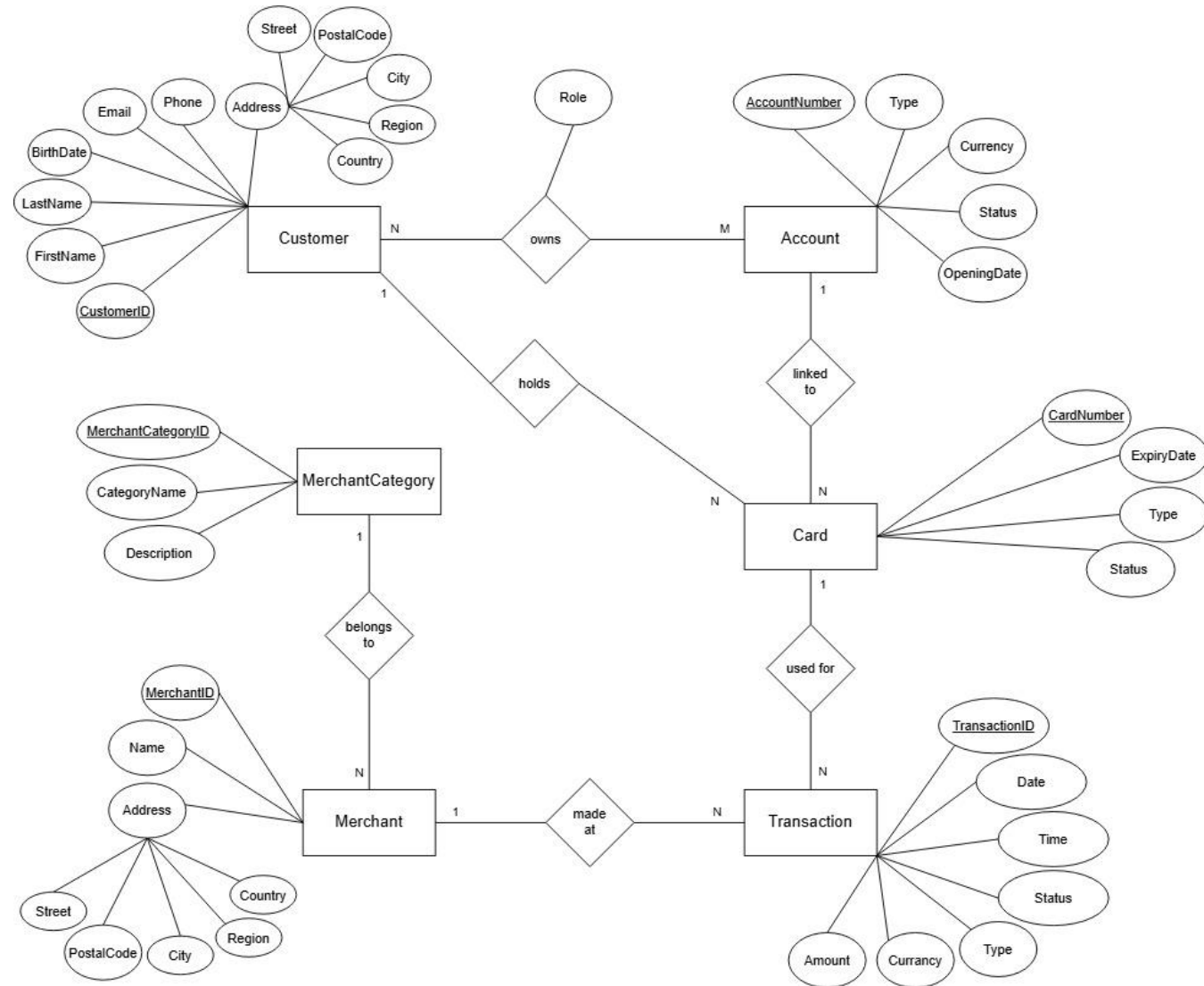
Key Business Rules

- Each card has one cardholder and is linked to one account
- Each transaction uses one card and is made at one merchant
- Each merchant belongs to one merchant category
- One customer/account/card can participate in many transactions
- Each Merchant represents one individual merchant location; merchant chains or brands are not modeled separately.



Business Database & Source Data

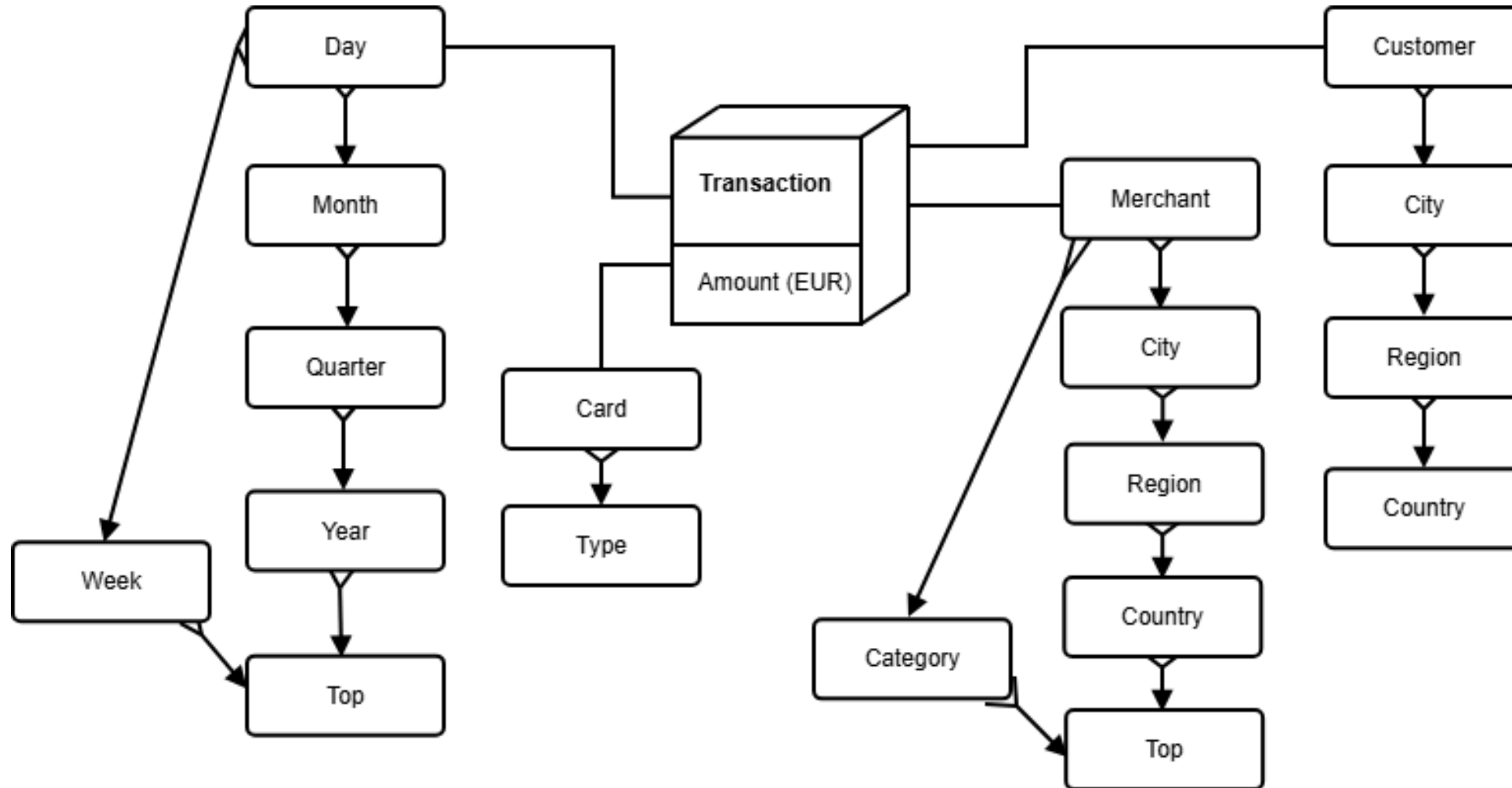
Business Database – ER Model



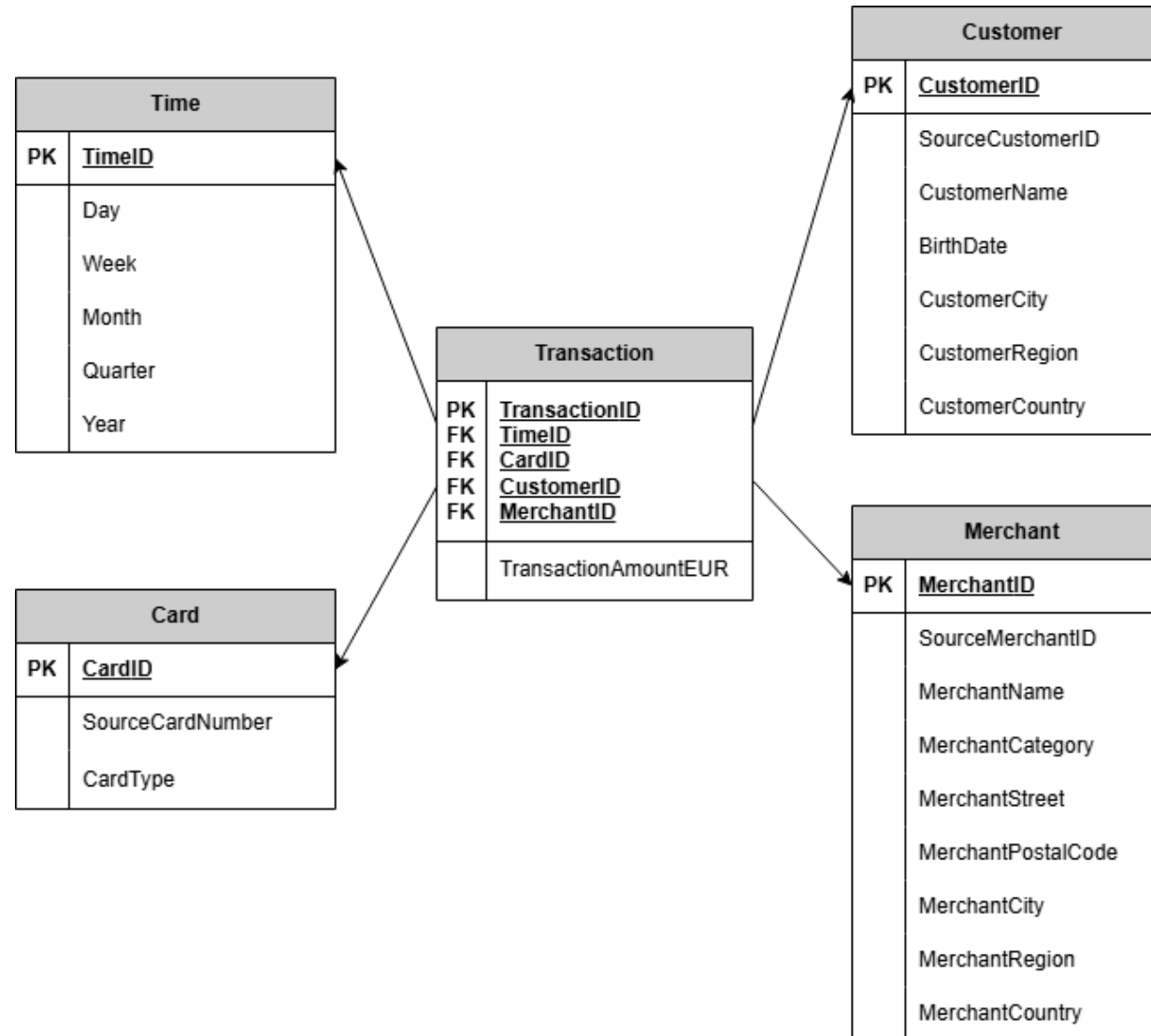
Source Data

- AI-generated banking dataset (GPT-5.6 Sol)
- 119 customers, 179 cards, 40 merchants, 12 merchant categories
- Around 2,000 card transactions
- Period: 2024-2026
- Currencies: EUR, USD, GBP, CHF

Multidimensional ER Model

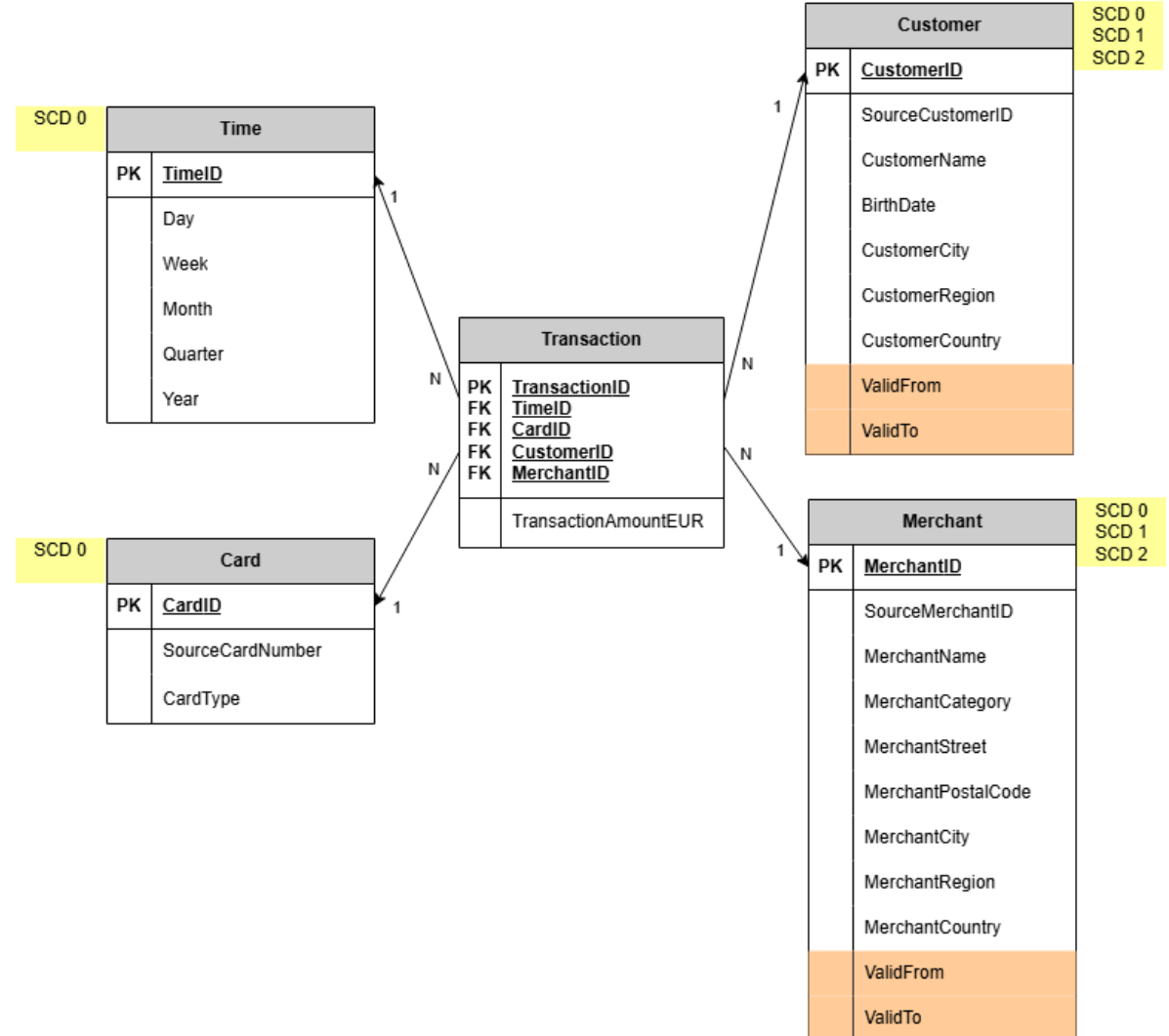


Logical Star Schema



Slowly Changing Dimensions (SCD)

- Time & Card: SCD Type 0
- Customer & Merchant: combination of SCD 0 / 1 / 2
- Type 2 history: ValidFrom / ValidTo



Top merchant category per region in 2026

	MerchantRegion	MerchantCategory	TransactionCount	TotalAmountEUR
1	Nordrhein-Westfalen	Electronics	22	11002.60
2	Île-de-France	Electronics	10	5747.94
3	Bremen	Electronics	7	4969.75
4	England	Travel & Transport	18	3192.51
5	Sachsen	Hotels & Lodging	7	2349.30

ETL Process & Scheduling

- ETL implemented with Stored Procedures
- Incremental Fact Load using control table / Last_loading_Date
- Currency conversion to EUR during transformation
- ETL validation: row counts, duplicates, foreign keys, future dates
- Planned schedule: daily batch; manual execution in project

Takeaways & Lessons Learned

Highlights

- Complete pipeline from relational Business DB to analytical Star Schema
- ETL with Stored Procedures, mapping and validation successfully implemented

Problems & Solutions

- Multi-currency transactions → converted to EUR during ETL
- No source change timestamps → initial SCD records use ValidFrom = 1900-01-01

Key Takeaway

- Source data completeness determines what can be reliably analyzed and modeled in the DWH
- Business rules and analytical requirements should be clear before modelling



Thank you

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